

Financial Anomalies for Indian Markets

Documentation

March 2026

1 Introduction

Estimating the equity risk premium is a topic of interest for academicians and investors alike. Over the years, various risk factors have been identified that have enhanced our understanding of variation in the equity returns. The most prominent set of risk factors has been proposed by Fama-French (Fama and French 1993) (Fama and French 2016). They identified Value, size, investment, and operating profitability to be the most significant features, which explain the stock price movements. (Carhart 1997) introduced momentum as another factor that is widely used by investors. Predicting stock returns is inherently a hard problem due to a low signal-to-noise ratio. Constructing portfolios negates the idiosyncratic changes of individual stocks, which helps in boosting the signal-to-noise ratio (Gu, Kelly, and Xiu 2020).

Despite its popularity, there is limited literature in the Indian context. The most notable work has been by (Agarwalla, Jacob, and Varma 2014) where they constructed Fama-French 3 factors and momentum for the stocks listed on the Bombay stock exchange. They have recently updated the universe of stocks to include both NSE and BSE stocks along with some additional filters. However, they produce only size, value and momentum factors which are updated thrice every year. Another issue with their approach is that they consider many small and not frequently traded firms as part of their universe. A more recent study by Kelly et al. constructed 153 factors across 13 themes for 93 countries. In order to remain consistent across countries, they apply a uniform methodology to produce these anomalies. A major drawback of this approach is that it undermines the market structure of each country and ignores the investability aspect of the stocks. Since a vast majority of the stocks listed on the Indian stock exchanges are too small and illiquid and the portfolio returns may not be representative of the market. In short, the existing sources suffer from either one or more of the following problems, first, inclusion of small and illiquid stocks for factor construction, secondly, limited coverage of factors, and lastly they are not updated monthly which can be used by investors. We propose a rigorous and empirically grounded approach to solve for these problems, our methodology ensures that the idiosyncrasies of the Indian equity markets are captured. We have extended the factor coverage by adding Investment and Operating Profitability. To make sure that the factor returns are reflective of the tradable universe in the Indian markets, we have performed careful analysis of the market structure before applying filters. The factor returns are provided at two levels at the entire universe after applying filters as well as top 300 stocks.

2 Data for Factor Construction

The firm-level balance sheet, daily stock market variables and share suspension data are sourced from the CMIE Prowess DX database. We extracted the daily stock market variable along with the annual balance sheets of publicly traded companies listed on the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). The time period for our analysis spans 25 years, from April 2000 to March 2024. There are a total of 7757 unique firms in our entire dataset.

2.1 Portfolio Formation

The entire factor construction has been organized under Portfolio Years. In our case, a Portfolio Year (PY) starts from October to September. Concretely, for a trading day t , we define the PY in the following manner.

$$PY(t) = \begin{cases} \text{year}(t) + 1, & \text{if } \text{month}(t) \in \{10, 11, 12\}, \\ \text{year}(t), & \text{if } \text{month}(t) \in \{1, \dots, 9\}. \end{cases}$$

The reason behind this convention is that a vast majority of the Indian companies have their financial end year in March and the financial results are often announced with a few months lag. Following the standard practice of Fama-French, we take a six month lag and form portfolios on end September. In contrast, JKP takes a four month lag instead of six. The breakpoint calculation for each factor and the holding universe for the next portfolio year is constructed using the data from the previous portfolio year.

2.2 Exchange selection

The Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE) are India's leading stock exchanges. The BSE was established in 1875 and has the largest number of listed companies. The NSE on the other hand was founded in 1992 to introduce modern trading practices and leads in terms volume and total market capitalization. Companies are often listed on both the exchanges and this creates a challenge of exchange selection for factor construction. In our base data, there are 4150 and 633 firms which exclusively listed on the BSE and NSE respectively. Although the number of firm exclusively listed on NSE are comparatively lower than BSE but there has been a sharp rise in recent years. There are 2974 companies which are listed on both the exchanges. The exchange decision is made annually based on aggregate traded value in the preceding Oct-Sep year (PY t), then applied to all daily rows in the holding year (PY $t+1$). The decision is shifted forward by one year to avoid look-ahead bias. We follow a simple rule of exchange selection, if a stock has been listed on the NSE then we take the NSE values else we choose BSE. This approach is line with the market trends as NSE has emerged as the dominant player in terms of volume.

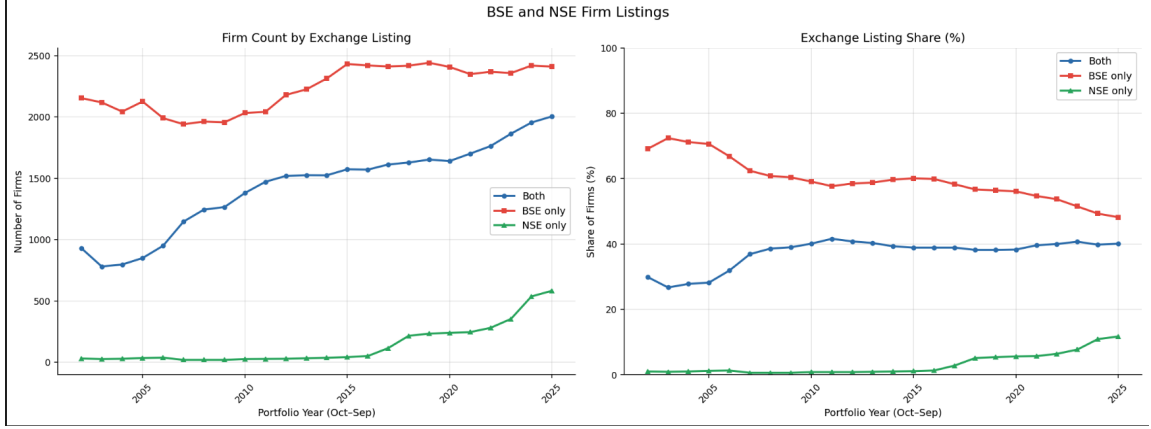


Figure 1: NSE and BSE listings over the years

3 Filters for Indian Equity Markets

3.1 Penny Stock Filter

3.1.1 Overview

We define a particular stock a “penny stock” if its median daily closing share price, in the preceding one year of the portfolio formation date is less than or equal to Rs. 10 . We remove all the penny stocks from our holding universe in the next Portfolio year in order to ensure the stocks used for factor construction are investable and don’t exhibit volatile with relatively small volumes of trades.

$$\tilde{P}_{i,PY} = \text{median}\{P_{i,t} : t \in PY\}, \quad \text{and} \quad \text{Penny Filtered}_{i,PY} = \mathbb{1}\{\tilde{P}_{i,PY} \geq 10\}.$$

$$\text{Eligible Non-Penny Stocks}_{i,PY+1} := \text{Penny Filtered}_{i,PY}.$$

3.1.2 Why do we need a penny stock filter?

Pandey & Sehgal (2016) argue that penny stocks are highly speculative and have limited liquidity which can lead to price manipulations. Additionally, such stocks show extreme volatility in price with large swings in either directions. These characteristics make such stocks both expensive to trade, as well as undesirable to use when tracking performance of macro variables like factors over time, since they could skew the sample.

Moreover, penny stocks have vastly lower oversight from retail and institutional investors, meaning that data collated about them is relatively less verified, and thus may contain a significant number of mistakes or aberrations.

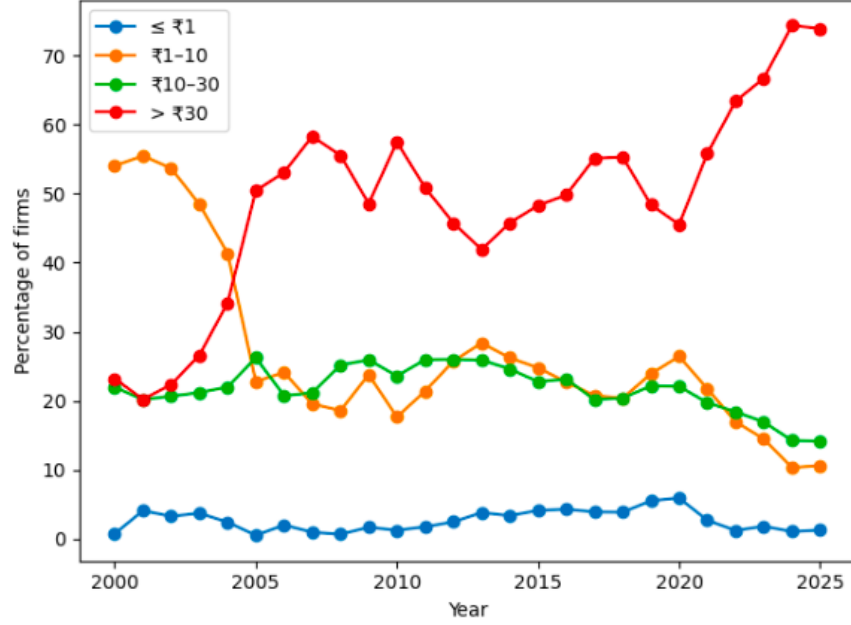


Figure 2: Percentage of firms in each price bucket, FY2000-FY2025

3.1.3 What is a good choice for the filter and why?

The U.S Securities and Exchange Commission (SEC) has an official definition of penny stocks, stock below \$5 share price are classified as penny stocks. However, the Securities and Exchange Board of India (SEBI) has not issued a similar definition for penny stocks in India or any minimum trading price which could have been used as a proxy. Several studies focusing on the Indian equity markets have defined penny stocks (Pandey & Sehgal (2016), Bhattacharyya & Chandra (2016)) uses a cutoff of Rs. 10. However, Agarwalla, S. K., Jacob, J. & Varma, J. R. (2013) use a penny stock threshold of Re. 1. (Jegadeesh & Titman (2023)) in their cross country study involving India has used a cutoff of \$1.

The percentage of firms with share price less than Rs. 10 remained relatively stable from FY2005 - FY2020 at 25-30% of the total firms. On the other hand, the percentage of firms with share price less than Rs. 30 dropped uniformly after FY2020, probably due to inflows into the equity markets and the resultant broad-based growth in market caps. The percentage of stocks with share price below Rs. 1 is extremely small- 0.5-3% across sample years, so Rs. 1 is too low a cutoff to use, as it will not significantly alleviate the three above mentioned problems which arise when considering penny stocks

Table 1 shows also that the 25th percentile of the annual share price distribution hovered around Rs. 10 from FY2005 till FY2020, post which it climbed to around Rs. 30.

Thus, given the stability of the Rs. 10 cutoff over the vast majority of the sample period and a cutoff of Rs. 1 captures very few firms, we believe that Rs. 10 is an appropriate ceiling to use in order to define a "penny stock".

Table 1: Percentile Distribution by Year

Year	p10	p25	p50	p90
2005	5.18	10.73	30.68	297.33
2010	5.42	13.89	42.31	347.69
2015	2.66	8.16	27.58	405.85
2020	1.91	6.60	24.08	390.20
2025	8.22	27.93	98.90	990.02

3.2 Microcap Filter

In addition to the penny stock filter, we also apply the microcap filter. The goal of this filter is to drop stocks with low market capitalization which are often harder to trade despite them not being a penny stock. The data shows a significant overlap between microcap and penny stocks. We follow the J.R. Varma’s approach to define the microcap cutoff which involves dropping all the firms whose market capitalization on the last day of PY is less than the 10 percent of the median market capitalization. **insert microcap cutoff image**

3.3 Liquidity Filter

The presence of a large number of illiquid firms in the market can potentially distort returns. Discontinuities in trading can stem from various multiple factors such as suspensions, lack of supply or demand for the shares. In order to ensure that the factor returns are not influenced by illiquid firms we drop illiquid firms from the factor calculation pipeline. This treatment ensure that the returns are representative of the tradable universe only.

We call a firm illiquid on the portfolio formation day if the firm has not traded every trading week atleast once. A trading week is defined as the week starting from Monday and ending on Friday. If a firm is listed during the year then the liquidity calculation is applied only from the week it started trading. There exists other approaches of applying the liquidity filter. Notably, JR varma et al. consider a firm illiquid if it has traded less than 50 days in a financial year. One of the problems with their criteria is that there can be cases where firms may appear to be liquid while exhibiting prolonged periods on liquidity. Our criteria is more stringent and considers only those companies which are highly liquid in nature.

4 Annual Factor Construction

4.1 Formulas & Definitions

4.1.1 Operating Profitability (OpProf)

A measure of profitability scaled by book equity. Note that it uses lagged Book Equity (from year $t - 1$) as the denominator.

$$\text{OpProf}_t = \frac{\text{Sales}_t - \text{COGS}_t - \text{S\&D Exp}_t - \text{Interest}_t}{\text{Book Equity}_{t-1}} \quad (1)$$

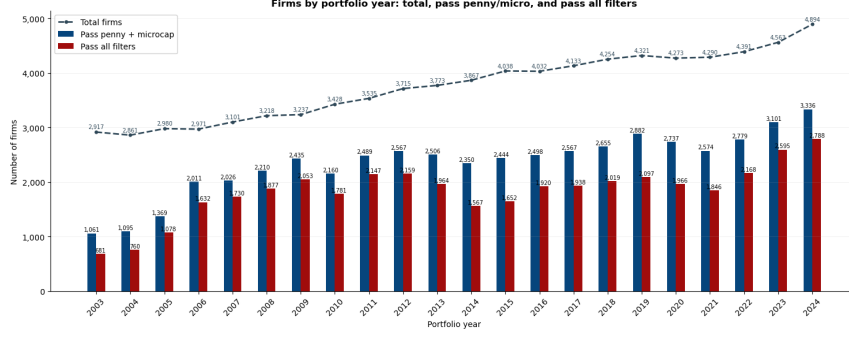


Figure 3: Firm distribution before and after applying filters

4.1.2 Investment (Inv)

Asset growth rate. The code calculates this using lags of total assets relative to the current row's financial year. High asset growth is considered "Aggressive" (bad signal), while low/negative growth is "Conservative" (good signal).

$$\text{Investment}_t = \frac{\text{Total Assets}_{t-1} - \text{Total Assets}_{t-2}}{\text{Total Assets}_{t-2}} \quad (2)$$

4.1.3 Book-to-Market (B/M)

Valuation ratio. Uses Book Equity and Market Cap aligned to the financial year end (typically March 31st).

$$\text{Book-to-Market}_t = \frac{\text{Book Equity}_t}{\text{Market Cap}_t} \quad (3)$$

We will later use this dataset for our factor returns calculations.

5 Monthly Factor Construction

5.1 Momentum (12-1 Month)

Momentum is calculated on a rolling monthly basis. It is defined as the cumulative return over the past 12 months, **excluding** the most recent month (to avoid short-term reversal effects).

- **Calculation:** Calculate monthly returns, shift the data by 2 months, and compute the rolling product over a window of 11 months.

$$\text{Momentum}_t = \prod_{i=2}^{12} (1 + R_{t-i}) \quad (4)$$

5.2 Size (Market Cap)

Logic: Uses the market capitalization from the last trading day of the **previous month**.

6 Portfolio Construction Methodology

6.1 Universe Definition

For every year Y :

1. Identify

6.2 Breakpoint Calculation (The Buckets)

Breakpoints are calculated annually (for fundamental factors) and monthly (for momentum/size) based **only on the Top 300 firms**.

Factor	Metric	Breakpoints	Labels
Op. Profitability	OpProf	30th / 70th	W (Weak), N (Neutral), R (Robust)
Investment	Asset Growth	30th / 70th	C (Conservative), N (Neutral), A (Aggressive)
Value	Book-to-Market	30th / 70th	G (Growth), N (Neutral), V (Value)
Momentum	12-1 Mo Ret	30th / 70th	L (Loser), N (Neutral), W (Winner)
Size	Market Cap	90th	S (Small), B (Big)

Table 2: Factor Breakpoints and Labels

6.3 Rebalancing Schedule

- **Fundamental Factors (Value, Inv, OpProf):** Signals calculated on financials from March 31, Year T . Portfolios formed on **October 1, Year T** and held until **September 30, Year $T + 1$** .
- **Momentum / Size:** Rebalanced **Monthly**.

7 Return Calculation Logic

Portfolios are intersected by **Size** and the **Factor** of interest (e.g., Small-Value, Big-Winner). For every factor, 6 portfolios are created (2 Size $\times$ 3 Factor Buckets).

7.1 Weighting Schemes

- **Value-Weighted (VW):**

$$w_{i,t} = \frac{\text{Market Cap}_{i,t-1}}{\sum \text{Market Cap}_{j,t-1}} \quad (5)$$

- **Equal-Weighted (EW):**

$$w_{i,t} = \frac{1}{N_t} \quad (6)$$

7.2 Performance Statistics

- **Total Cumulative Geometric Return:** Calculates the total compounded growth over the entire period.

$$\text{Total Geo} = \left(\prod_{t=1}^N R_t \right) - 1$$

- **Annualized Geometric Return (CAGR):** The geometric mean return annualized over Y years.

$$\text{Ann. Geo} = \left(\prod_{t=1}^N R_t \right)^{\frac{1}{Y}} - 1$$

- **Total Arithmetic Return:** The simple sum of all monthly excess returns (not compounded).

$$\text{Total Arith} = \sum_{t=1}^N (R_t - 1)$$

- **Annualized Arithmetic Average Return:** The mean monthly return multiplied by 12.

$$\text{Avg. Arith} = \left(\frac{1}{N} \sum_{t=1}^N (R_t - 1) \right) \times 12$$

8 Factor Performance

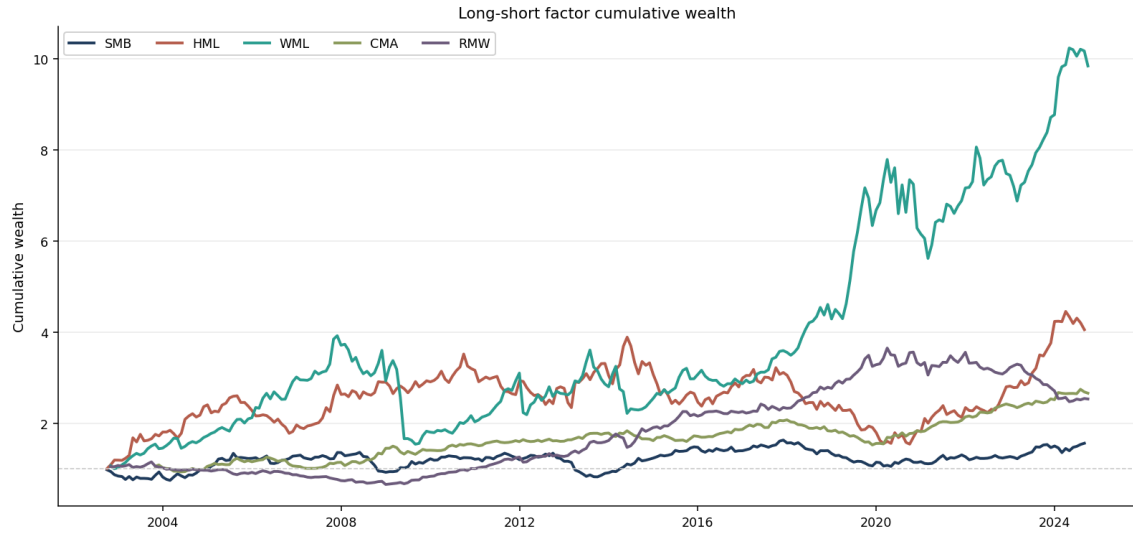


Figure 4: Cumulative Wealth from Long-Short factor returns

9 Factor breakpoints and Returns

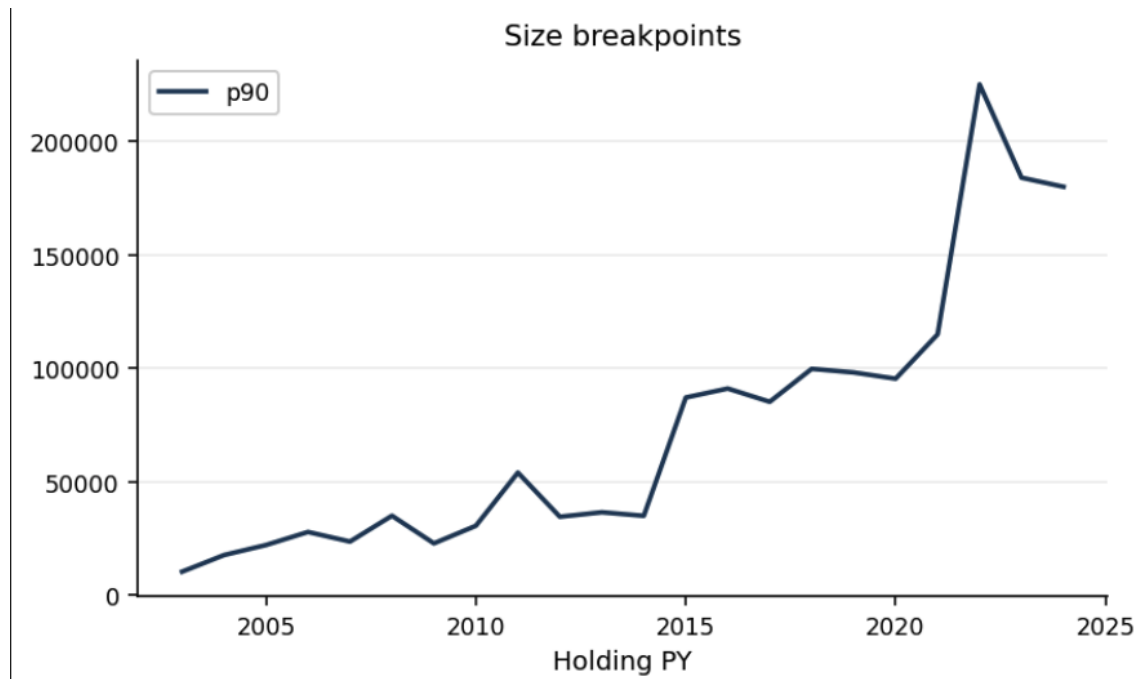


Figure 5: Size Breakpoint (in Millions)

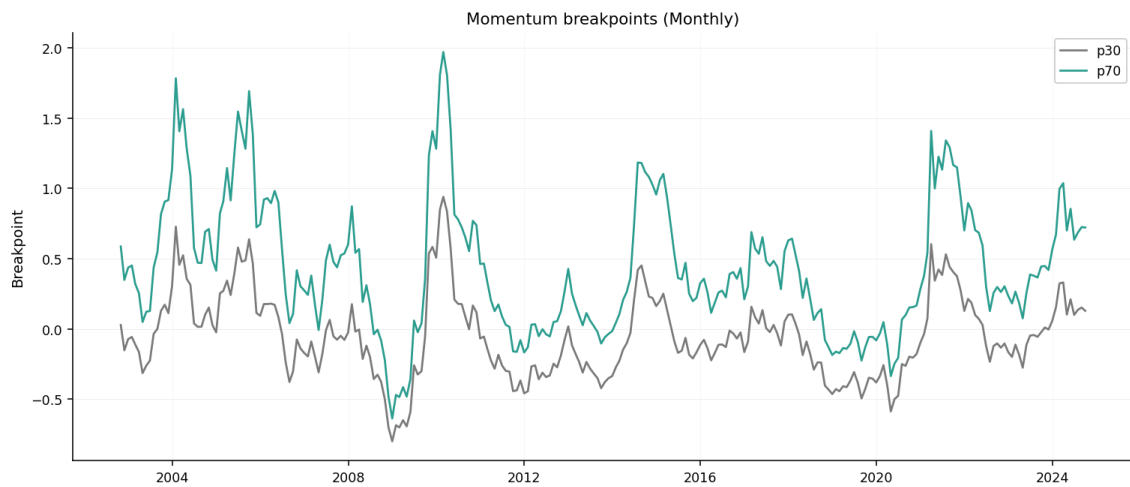


Figure 6: Momentum Breakpoint

10 Conclusion

$$I_{i,t} := \mathbb{1}\{r_{i,t} \text{ is observed}\}.$$

Let $\mathcal{W}_{PY}$ denote the set of complete 5-trading-day weeks in portfolio year PY .

% Market calendar weeks in portfolio year PY Let W_{PY} denote the number of complete (5-trading-day) weeks in portfolio year PY .

Let $I_{i,w} := \mathbb{1}\{\text{firm } i \text{ has at least one trade in week } w\}$.

$$A_{i,PY} := \sum_{w=1}^{W_{PY}} I_{i,w}.$$

Let $W_{i,PY}^{\text{IPO}}$ be the number of complete weeks in PY that occur on or after firm i 's listing date. Equivalently, restrict the sum to weeks w with trading days $t \geq t_i^{\text{list}}$:

$$A_{i,PY}^{\text{IPO}} := \sum_{w=1}^{W_{i,PY}^{\text{IPO}}} I_{i,w}.$$

$$\text{PassLiq}_{i,PY} := \mathbb{1}\left\{A_{i,PY}^{\text{IPO}} \geq 0.8 W_{i,PY}^{\text{IPO}}\right\}.$$

$$\text{EligibleLiq}_{i,PY+1} := \text{PassLiq}_{i,PY}.$$

]

$$\text{EligibleLiq}_{i,PY+1} := \mathbb{1}\{\text{Gap}_{i,PY} = 0\}.$$

References

- Agarwalla, Sobhesh Kumar, Joshy Jacob, and Jayanth Rama Varma (2014). “Four factor model in Indian equities market”. In: *Indian Institute of Management, Ahmedabad Working Paper* 2013-09, p. 05.
- Carhart, Mark M (1997). “On persistence in mutual fund performance”. In: *The Journal of finance* 52.1, pp. 57–82.
- Fama, Eugene F and Kenneth R French (1993). “Common risk factors in the returns on stocks and bonds”. In: *Journal of financial economics* 33.1, pp. 3–56.
- (2016). “Dissecting anomalies with a five-factor model”. In: *The Review of Financial Studies* 29.1, pp. 69–103.
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu (2020). “Empirical asset pricing via machine learning”. In: *The Review of Financial Studies* 33.5, pp. 2223–2273.